

SOLST01 Long Assessment 1

Solid State Physics

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1 Problem 1: Quantum Harmonic Oscillator

1.1 Analytical methodology

1.1.1 Time-Independent Schrödinger Equation

The time-independent Schrödinger Equation is

$$H |\psi\rangle = E |\psi\rangle$$

where the Hamiltonian H is

$$H = T + V$$

T and V are the kinetic and potential operators, respectively.

1.1.2 Quantum Harmonic Oscillator Hamiltonian

The Hamiltonian of the quantum harmonic oscillator is

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

where p and x are the momentum and position operators, respectively.

1.1.3 Projection onto 1D Position Space

When projected onto one dimension of position space, the time-independent Schrödinger equation becomes

$$\begin{aligned} H\psi(x) &= E\psi(x) \\ -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x)\psi(x) &= E\psi(x) \end{aligned}$$

1.1.4 Resulting Differential Equation

The time-independent Schrödinger equation is a differential equation:

$$\frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2m(E - V(x))}{\hbar^2} \psi(x) = 0$$

1.1.5 Analytical Energy Spectrum

The eigenvalues of H form a spectrum of values called the *energy spectrum* of the system described by H . For the quantum harmonic oscillator, the energy spectrum is [1, page 51]:

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

1.1.6 Analytical Energy Spectrum

The eigenwavefunctions of the quantum harmonic oscillator turn out to be Hermite polynomials and are given normalized by Griffiths and Schroeter [1, page 52] as

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}, \quad \text{where } \xi = \sqrt{\frac{m\omega}{\hbar}} x$$

and the first few H_n are:

$$\begin{aligned} H_0 &= 1 \\ H_1 &= 2\xi \\ H_2 &= 4\xi^2 - 2 \\ H_3 &= 8\xi^3 - 12\xi \\ H_4 &= 16\xi^4 - 48\xi^2 + 12 \\ H_5 &= 32\xi^5 - 160\xi^3 + 120\xi \end{aligned}$$

1.2 Numerical methodology

1.2.1 Finite Computational Interval, Grid Definition, and Grid Spacing

A position grid of N points is constructed as follows

$$x_i = x_{\min} + i\Delta x, \quad i = 0, 1, 2, \dots, N-1$$

where the grid spacing Δx is

$$\Delta x = \frac{x_{\max} - x_{\min}}{N-1}$$

1.2.2 Finite Difference Approximation to the 2nd Derivative in the Differential Equation

Let $\psi(x)$ be defined on a continuous interval of x and have continuous 1st and 2nd derivatives. The 2nd derivative, $\partial^2\psi(x)/\partial x^2$, at any interior point a of the interval may be approximated as

$$\left(\frac{\partial^2\psi(x)}{\partial x^2}\right)_{x=a} \approx \frac{\psi(a + \Delta x) + \psi(a - \Delta x) - 2\psi(a)}{(\Delta x)^2}$$

On our discrete grid, $\psi(x)$ becomes discrete $\psi(x_i)$ where $i = 0, 1, 2, \dots, N-1$ and its 2nd derivative becomes

$$\left(\frac{\partial^2\psi(x)}{\partial x^2}\right)_{x=x_i} \approx \frac{\psi(x_{i+1}) + \psi(x_{i-1}) - 2\psi(x_i)}{(\Delta x)^2}$$

that is, it is calculated from the wavefunction values at x_{i-1} , x_i and x_{i+1} .

1.2.3 Construction of the Kinetic Energy Matrix

Thus the kinetic energy operator at each point x_i is

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_i} \approx -\frac{\hbar^2}{2m} \frac{\psi(x_{i+1}) + \psi(x_{i-1}) - 2\psi(x_i)}{(\Delta x)^2}$$

Noting that the 2nd derivative is defined only at interior points $x_1, \dots, x_{N-2}$ and taking $\psi(x_0) = \psi(x_{\min}) = 0$ and $\psi(x_{N-1}) = \psi(x_{\max}) = 0$, we get the kinetic energy operator at each interior point:

$$\begin{aligned} -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_1} &\approx -\frac{\hbar^2}{2m} \frac{\psi(x_2) - 2\psi(x_1)}{(\Delta x)^2} \\ -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_2} &\approx -\frac{\hbar^2}{2m} \frac{\psi(x_3) + \psi(x_1) - 2\psi(x_2)}{(\Delta x)^2} \\ -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_3} &\approx -\frac{\hbar^2}{2m} \frac{\psi(x_4) + \psi(x_2) - 2\psi(x_3)}{(\Delta x)^2} \\ &\vdots \\ -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_{N-3}} &\approx -\frac{\hbar^2}{2m} \frac{\psi(x_{N-2}) + \psi(x_{N-4}) - 2\psi(x_{N-3})}{(\Delta x)^2} \\ -\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_{N-2}} &\approx -\frac{\hbar^2}{2m} \frac{\psi(x_{N-3}) - 2\psi(x_{N-2})}{(\Delta x)^2} \end{aligned}$$

In matrix-vector form, this system of equations is

$$-\frac{\hbar^2}{2m} \begin{bmatrix} \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_1} \\ \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_2} \\ \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_3} \\ \vdots \\ \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_{N-3}} \\ \left(\frac{\partial^2 \psi(x)}{\partial x^2} \right)_{x=x_{N-2}} \end{bmatrix} \approx -\frac{\hbar^2}{2m(\Delta x)^2} \begin{bmatrix} -2 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & & & & \ddots & & & \\ 0 & 0 & 0 & 0 & \cdots & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} \psi(x_1) \\ \psi(x_2) \\ \psi(x_3) \\ \vdots \\ \psi(x_{N-3}) \\ \psi(x_{N-2}) \end{bmatrix}$$

1.2.4 Construction of the Potential Energy Matrix

The potential operator at each interior x_i is

$$V(x_i)\psi(x_i) = \frac{1}{2}m\omega^2 x_i^2$$

We may find the potential operator at every interior x_i :

$$\begin{aligned}
V(x_1)\psi(x_1) &= \frac{1}{2}m\omega^2 x_1^2 \psi(x_1) \\
V(x_2)\psi(x_2) &= \frac{1}{2}m\omega^2 x_2^2 \psi(x_2) \\
V(x_3)\psi(x_3) &= \frac{1}{2}m\omega^2 x_3^2 \psi(x_3) \\
&\vdots \\
V(x_{N-3})\psi(x_{N-3}) &= \frac{1}{2}m\omega^2 x_{N-3}^2 \psi(x_{N-3}) \\
V(x_{N-2})\psi(x_{N-2}) &= \frac{1}{2}m\omega^2 x_{N-2}^2 \psi(x_{N-2})
\end{aligned}$$

In matrix-vector form, this system of equations is

$$\begin{bmatrix} V(x_1)\psi(x_1) \\ V(x_2)\psi(x_2) \\ V(x_3)\psi(x_3) \\ \vdots \\ V(x_{N-3})\psi(x_{N-3}) \\ V(x_{N-2})\psi(x_{N-2}) \end{bmatrix} = \frac{1}{2}m\omega^2 \begin{bmatrix} x_1^2 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & x_2^2 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & x_3^2 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & & & & \ddots & & & \\ 0 & 0 & 0 & 0 & \cdots & 0 & x_{N-3}^2 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & x_{N-2}^2 \end{bmatrix} \begin{bmatrix} \psi(x_1) \\ \psi(x_2) \\ \psi(x_3) \\ \vdots \\ \psi(x_{N-3}) \\ \psi(x_{N-2}) \end{bmatrix}$$

1.2.5 Boundary Condition Implementation

We restate the implemented boundary conditions for clarity

$$\begin{aligned}
\psi(x_0) &= \psi(x_{\min}) = 0 \\
\psi(x_{N-1}) &= \psi(x_{\max}) = 0
\end{aligned}$$

In the wavefunction ket $|\psi\rangle$, these wavefunction values do not appear explicitly and are instead implemented by absence.

1.2.6 Normalization of Numerical Eigenvectors

An arbitrary wavefunction $\langle x|\psi\rangle = \psi(x)$ is not necessarily normalized, i.e. $\alpha \in \mathbb{R}$ is not necessarily 1 in

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = \alpha$$

However, dividing both sides by α

$$\begin{aligned}
\frac{1}{\alpha} \int_{-\infty}^{\infty} |\psi(x)|^2 dx &= 1 \\
\frac{1}{\alpha} \int_{-\infty}^{\infty} \psi(x)^* \psi(x) dx &= 1 \\
\int_{-\infty}^{\infty} \left[\frac{\psi(x)}{\sqrt{\alpha}} \right]^* \left[\frac{\psi(x)}{\sqrt{\alpha}} \right] dx &= 1
\end{aligned}$$

On our discrete grid, these become

$$\sum_{i=i}^{N-2} |\psi(x)|^2 \Delta x = \alpha$$

and

$$\sum_{i=i}^{N-2} \left[\frac{\psi(x_i)}{\sqrt{\alpha}} \right]^* \left[\frac{\psi(x_i)}{\sqrt{\alpha}} \right] \Delta x = 1$$

We may observe that

$$\psi_{\text{new}}(x) = \frac{\psi(x)}{\sqrt{\alpha}} \quad \text{and} \quad \psi_{\text{new}}(x_i) = \frac{\psi(x_i)}{\sqrt{\alpha}}$$

are normalized in their respective cases. It is assumed that $\alpha \neq 0$.

1.2.7 Eigenenergy Error Calculations

The formula

$$\epsilon_{E,n} = \left| \frac{E_n^{\text{num}} - E_n^{\text{exact}}}{E_n^{\text{exact}}} \right|$$

gives the relative error between the numerical and analytical eigenenergies at n and is used throughout this section.

1.2.8 Unit Scheme and Conversion Constants Used in the Computation

All constants, computations, and displayed values were performed or are in base SI units unless otherwise indicated.

1.2.9 Python Libraries Used

The eigenenergies of the quantum harmonic oscillator Hamiltonian $H = T + V$, where the forms of T and V are given in the sections “*Construction of the Kinetic Energy Matrix*” and “*Construction of the Potential Energy Matrix*” respectively, are calculated using the `linalg.eig()` method of the Python library `numpy`. Then, they were arranged in ascending order, assigned numbers $n = 0, 1, 2, \dots$ in that order, and the 1st twenty (20) were plotted using the Python library `matplotlib`.

1.3 Results

1.3.1 Quantum Harmonic Oscillator Potential

The potential of the quantum harmonic oscillator described by Table (1) over the discrete position grid x_i is shown in in Fig. (1). It is a parabola centered on $x = 0$.

1.3.2 Numerical and Analytical Eigenenergies

The 1st twenty (20) numerical eigenenergies of the quantum harmonic oscillator system described by Table (1) are shown on Fig. (2). As a check, the 1st twenty (20) analytical eigenenergies are shown on the same figure. The numerical eigenenergies show strong agreement with their analytical counterparts.

The eigenenergies plotted on Fig. (2) are numerically listed in Table (2) along with their corresponding relative errors. Relative error is observed to increase with increasing n .

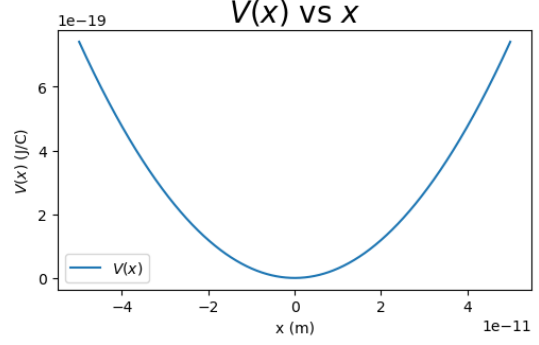


Figure 1: Harmonic Oscillator Potential

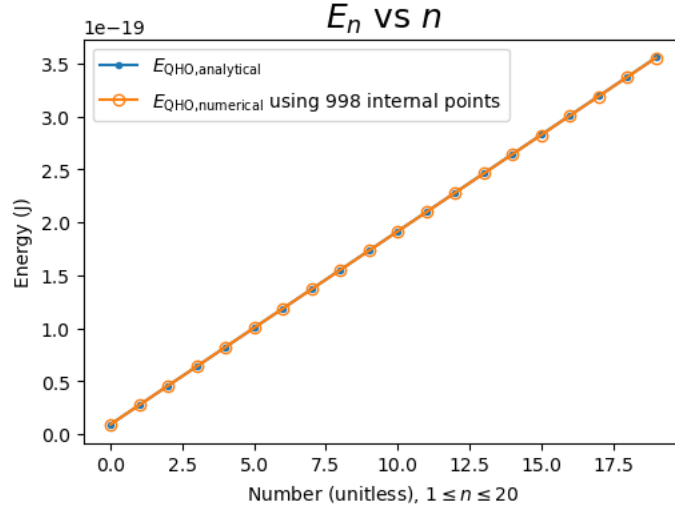


Figure 2: Numerical and Analytical Eigenenergies vs. n

System Parameter	Value	Notes
Start of Well	5×10^{-11} m	-
End of Well	-5×10^{-11} m	-
Total No. of Grid Points	1000	-
Mass of Trapped Particle	1.99×10^{-26} kg	Mass of One (1) Carbon Atom
Oscillator Frequency	1.728×10^{12} s ⁻¹	Einstein Frequency of Diamond, from Simon [2, page 13]

Table 1: System Parameters for Figs. (1,2)

n	E_n^{num} (eV)	E_n^{exact} (eV)	$\Delta E_n \times 10^6$ (eV)	$\epsilon_{E,n} \times 10^6$
0	0.057	0.057	-1.2	20.4
1	0.171	0.171	-5.8	34.1
2	0.284	0.284	-15.1	53.2
3	0.398	0.398	-29.1	73.0
4	0.512	0.512	-47.7	93.2
5	0.626	0.626	-71.0	113.4
6	0.740	0.740	-98.9	133.7
7	0.853	0.853	-131.5	154.1
8	0.967	0.967	-168.7	174.4
9	1.081	1.081	-210.6	194.8
10	1.195	1.195	-257.2	215.2
11	1.308	1.309	-308.4	235.6
12	1.422	1.422	-364.2	256.1
13	1.536	1.536	-424.8	276.5
14	1.650	1.650	-489.9	296.9
15	1.763	1.764	-559.8	317.4
16	1.877	1.878	-634.3	337.8
17	1.991	1.991	-713.4	358.2
18	2.104	2.105	-797.2	378.7
19	2.218	2.219	-885.7	399.1

Table 2: Relative Errors for Fig. (2) Eigenenergies

1.3.3 1st Three Numerical Probability Densities

The 1st three (3) normalized probability densities, namely $P_n(x)$ where $n = 0, 1, 2$, of the quantum harmonic oscillator may be seen in Fig. (3). The general shape of these probability densities is as expected: $P_0(x)$ has no nodes, $P_1(x)$ has one node, and $P_2(x)$ has two nodes. Furthermore, it is also expected that probability peaks closer to the center $x = 0$ are shorter than those farther away from the center; this pattern starts to be observed in $P_2(x)$, where the central probability peak is lower than the left and right probability peaks.

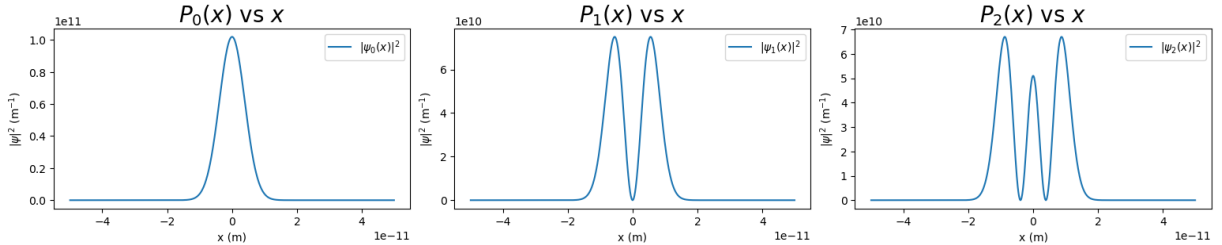


Figure 3: Probability Distributions for $n = 0, 1, 2$

1.3.4 1st Three Wavefunctions

The 1st three (3) normalized numerical wavefunctions and the 1st three (3) normalized analytical wavefunctions, namely $P_n(x)$ where $n = 0, 1, 2$, of the quantum harmonic oscillator may be seen in Fig. (4).

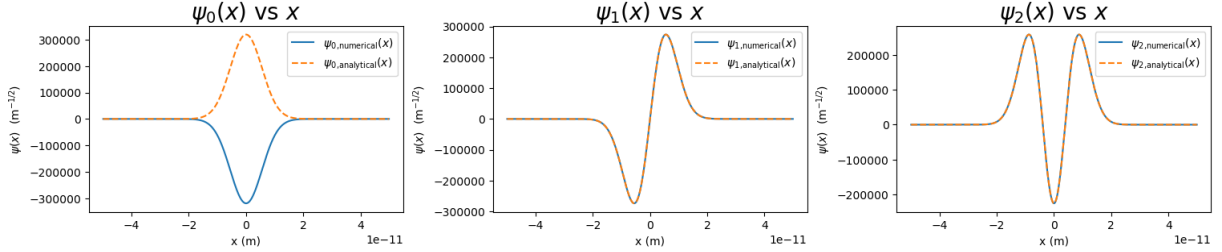


Figure 4: Wavefunctions for $n = 0, 1, 2$

1.3.5 $n = 10$ Probability Density

Thought this probability density function looked cool.

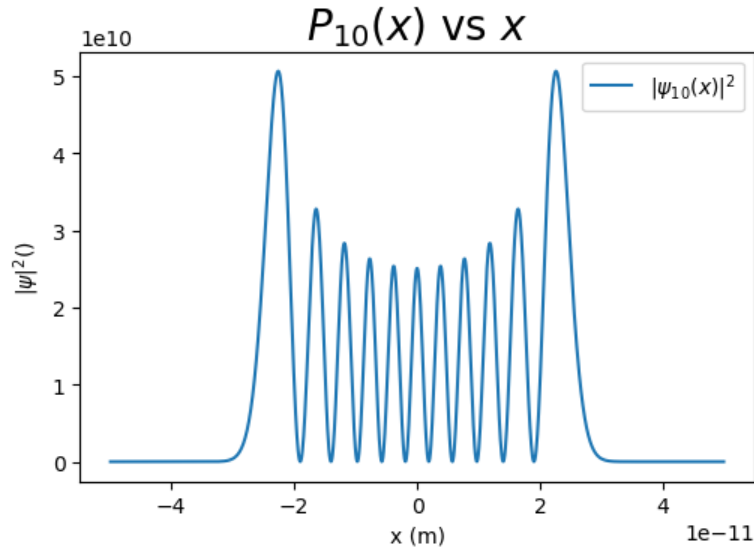


Figure 5: Wavefunction for $n = 10$

1.3.6 Sensitivity to No. of Grid Points

The relative errors for eigenenergies $n = 0, 1, 2, 3, 4$ as a function of no. of grid points are shown in Table (3). Relative error is observed to decrease with increasing no. of grid points.

No. of Grid Points	$\epsilon_{E,0} \times 10^3$	$\epsilon_{E,1} \times 10^3$	$\epsilon_{E,2} \times 10^3$	$\epsilon_{E,3} \times 10^3$	$\epsilon_{E,4} \times 10^3$
30	24.90	42.47	68.15	97.05	128.21
50	8.57	14.39	22.64	31.38	40.41
100	2.09	3.48	5.44	7.49	9.58
500	0.08	0.14	0.21	0.29	0.37
1000	0.02	0.03	0.05	0.07	0.09

Table 3: Grid Point Sensitivity of Relative Error. Other system parameters (Table 1) remain fixed.

1.3.7 Sensitivity to Grid Size

The relative errors for eigenenergies $n = 0, 1, 2, 3, 4$ as a function of grid size are shown in Table (4). Relative error is generally observed to increase with increasing deviation from the original half-width ℓ , with the increase being more drastic for decreasing half-widths.

Well Half-Width	$\epsilon_{E,0} \times 10^3$	$\epsilon_{E,1} \times 10^3$	$\epsilon_{E,2} \times 10^3$	$\epsilon_{E,3} \times 10^3$	$\epsilon_{E,4} \times 10^3$
$(1/8)\ell$	1098.92	1698.93	2561.73	3480.53	4420.08
$(1/5)\ell$	145.38	302.67	564.11	878.31	1218.14
$\ell = 5 \times 10^{-11}$	0.02	0.03	0.05	0.07	0.09
5ℓ	0.51	0.85	1.33	1.83	2.33
8ℓ	1.31	2.19	3.41	4.70	6.00

Table 4: Grid Size Sensitivity of Relative Error. Other system parameters (Table 1) remain fixed.

1.3.8 Average Energy and Heat Capacity

The numerical average energy and numerical Einstein heat capacity of one (1) quantum harmonic oscillator are shown as functions of temperature in Fig. (6).

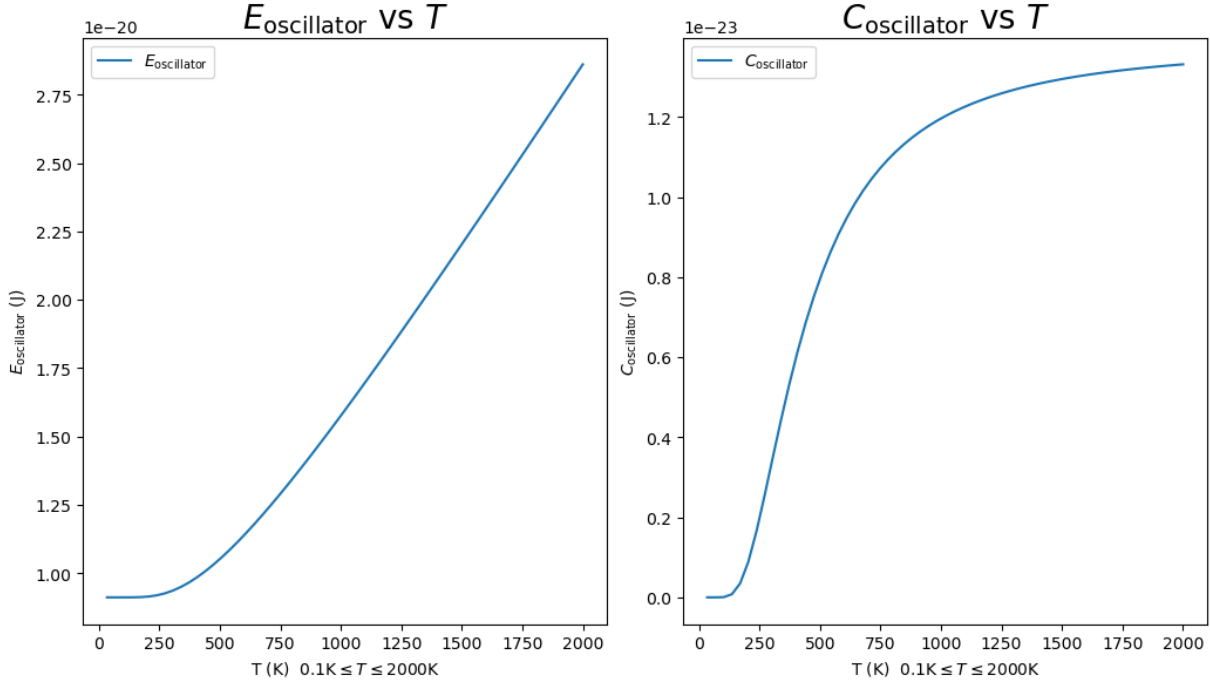


Figure 6: Mean Internal Energy (left) and Heat Capacity (right) vs Temperature for One (1) Quantum Harmonic Oscillator

1.4 Discussion

The connection between Dirac notation and the position space Schrödinger equation, or equivalently the connection between the finite-grid and continuous interval problems, is elaborated upon in the subsection “*Numerical Methodology*.”

The numerical eigenenergies were checked against the analytical eigenenergies by calculating their relative errors $\epsilon_{E,n}$ using the formula given in “*Eigenenergy Error Calculations*.”

The quantum harmonic oscillator modeled in this section is more accurately a quantum harmonic oscillator placed inside an infinite potential well with Dirichlet boundary conditions $\psi = 0$. This means that the problem being solved is a fundamentally different problem and that the numerical eigenenergies cannot come arbitrarily close to the analytical eigenenergies.

Decreasing the grid spacing Δx decreases the accuracy of the numerical eigenenergies since this decreases the total number of eigenenergies that can be obtained from diagonalizing the Hamiltonian. Increasing or decreasing the well half-width ℓ also decreases the accuracy of the numerical eigenenergies since increasing it decreases the number of grid points located within the turning points of the classical counterpart of the quantum harmonic oscillator, while decreasing it puts the walls of the infinite potential well close to or within the classical turning points, respectively. High energy states are more susceptible to changes in ℓ than low energy states because the classical turning

points of the former are farther away from the center $x = 0$ and the corresponding wavefunctions occupy more space inside the infinite potential well.

The energy level weighting method and Bose-Einstein occupation term are equivalent since the independent oscillators are bosons with respect to each other, i.e. any two oscillators in the system can be in the same state and have the same energy at any time. However, the energy level weighting method can give average energies arbitrarily close to those calculated using the Bose-Einstein occupation term only if the former can accomodate infinitely many and infinitely large eigenenergies.

In this section, eigenenergies are displayed in terms of state number n and not in terms of wavevector k as they are for some other systems such as the free particle. This is because the average momentum, and therefore the average wavevector, of particles in infinite potential wells and/or quadratic potentials is 0, regardless of the energy of the particle.

2 Problem 2: Einstein Solid

2.1 Results

2.1.1 Numerical and Analytical Heat Capacity

The numerical and analytical Einstein heat capacities are shown in Fig. (7). Across the temperature range 0.1 to 2000 K, the numerical heat capacities show strong agreement with their analytical counterparts.

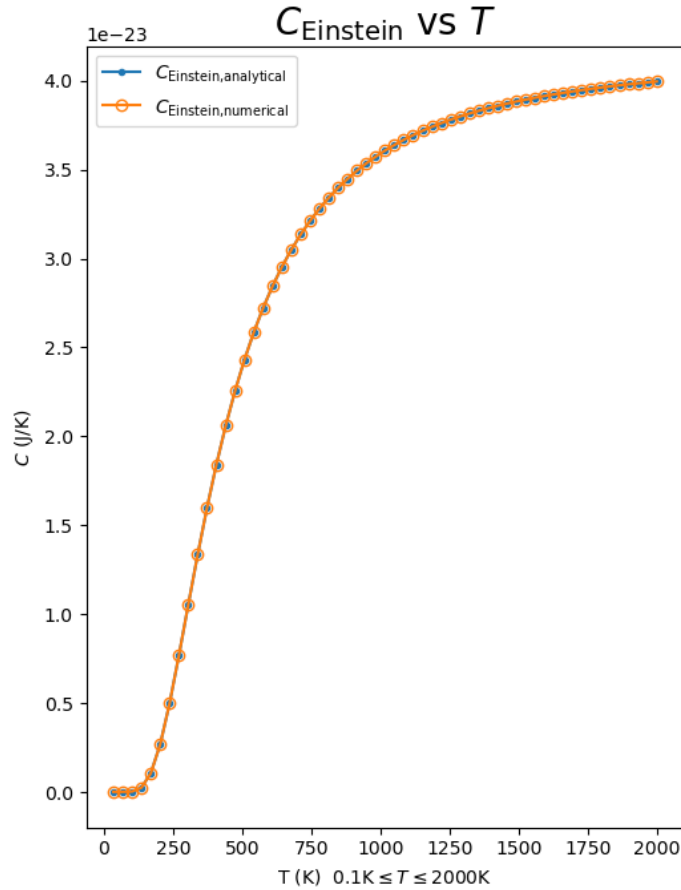


Figure 7: Numerical and Analytical Einstein Heat Capacity vs Temperature

System Parameter	Value	Notes
No. of Atom(s)	1	-
Einstein Frequency	$1.728 \times 10^{14} \text{ s}^{-1}$	for Diamond, from Simon [2, page 13]
Finite Difference Magnitude (ΔT)	0.001 K	arbitrary

Table 5: System Parameters for Einstein Heat Capacity

2.2 Discussion

The Einstein solid modeled in this section takes in the energy spectrum of the quantum harmonic oscillator modeled in the previous section as its input for calculating internal energy and heat capacity.

3 Problem 3: Debye Solid

3.1 Analytical Methodology

The Debye heat capacity is given by Simon [2, page 13] as

$$C(T) = N_{\text{atom}} k_B \frac{(k_B T)^3}{(\hbar \omega_{\text{Debye}})^3} \frac{12\pi^4}{5}$$

3.2 Results

3.2.1 Numerical and Analytical Heat Capacity

The numerical and analytical Debye heat capacities are shown in Fig. (8). Across the temperature range 0.1 to 200 K, the numerical heat capacities show strong agreement with their analytical counterparts, but with decreasing agreement closer to 200 K.

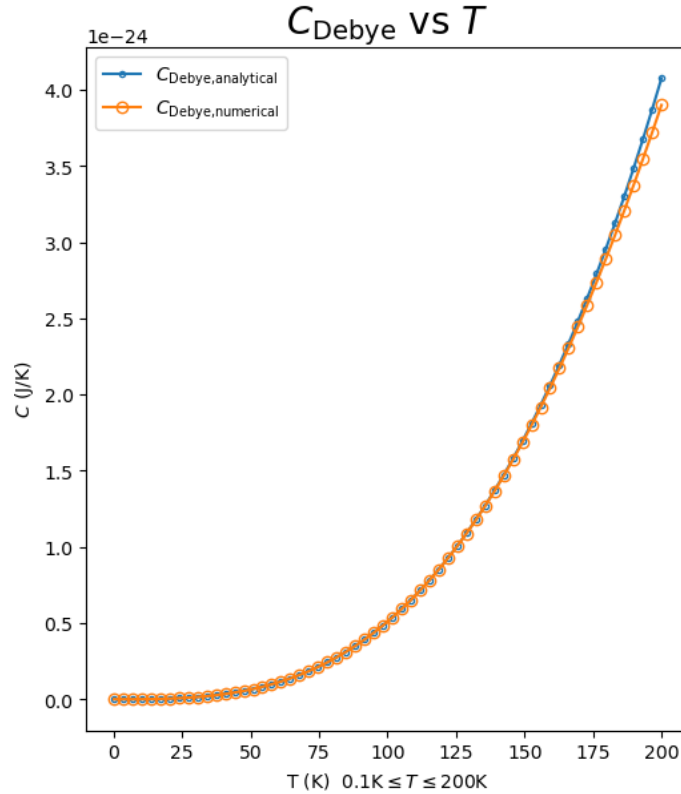


Figure 8: Numerical and Analytical Debye Heat Capacity vs Temperature

System Parameter	Value	Notes
No. of Atom(s)	1	-
Debye Frequency	$2.422 \times 10^{14} \text{ s}^{-1}$	for Diamond, from Simon [2, page 13]
Finite Difference Magnitude (ΔT)	0.001 K	arbitrary

Table 6: System Parameters for Fig. (8)

3.2.2 Einstein and Debye Heat Capacities Compared

The previously-obtained numerical and analytical Einstein and Debye heat capacities are shown together in Fig. (9). As before, the numerical and analytical Einstein heat capacities show strong agreement with each other across the temperature range 0.1 K to 2000 K. The numerical Debye heat capacities are greater than the Einstein heat capacities at lower temperatures, and are less than the Einstein heat capacities at higher temperatures, with the transition point being located around 300 K. The analytical Debye heat capacities begin to deviate significantly from the other three at around 250 K.

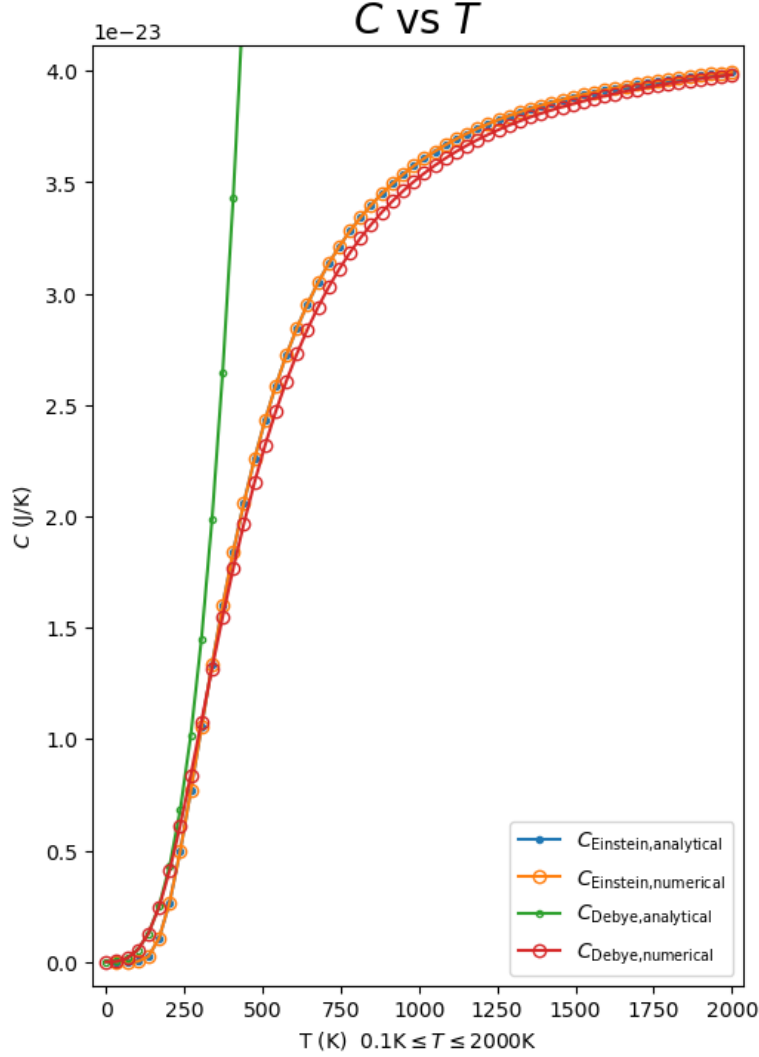


Figure 9: Einstein and Debye Heat Capacity vs Temperature

3.3 Discussion

The Debye model is an improvement on the Einstein model in that the former takes into account the fact that one atom may take on many different vibrational frequencies (normal modes) and not just one, the Einstein frequency.

4 Problem 4: Diatomic Linear Chain

4.1 Results

4.1.1 Numerical and Analytical Dispersion - Reduced Zone Scheme

The numerical and analytical dispersions are shown in Fig. (10) in the reduced zone scheme, i.e. all frequencies are placed within the 1st Brillouin zone. The upper branch is the optical branch, and the lower branch is the acoustic branch.

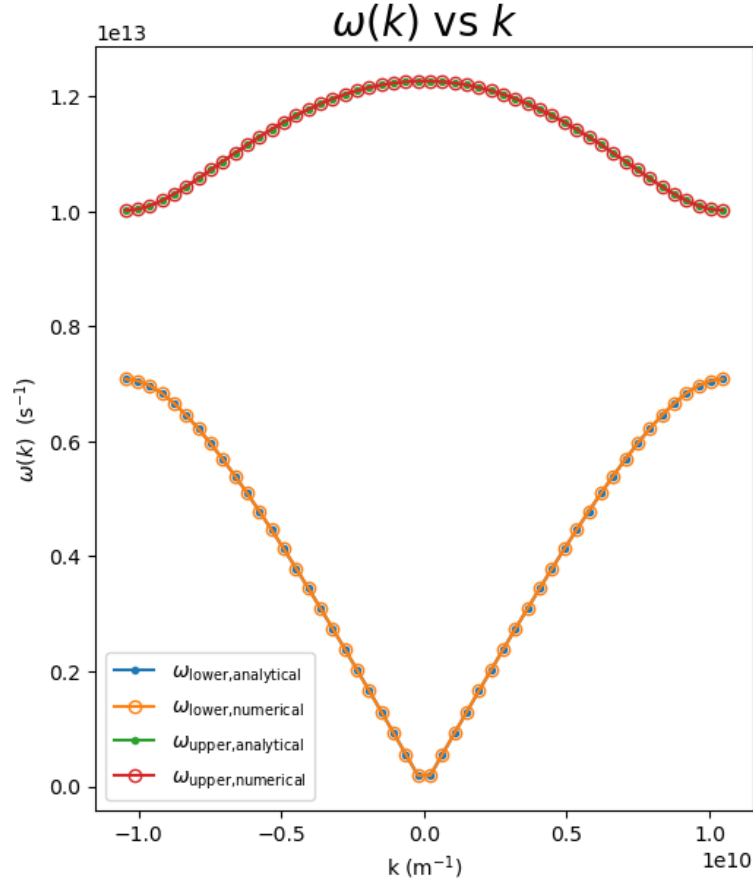


Figure 10: Reduced Zone Scheme for a Diatomic Linear Chain

System Parameter	Value	Notes
Mass 1	1.99×10^{-26} kg	Mass of One (1) Carbon Atom
Mass 2	$2(1.99 \times 10^{-26})$ kg	Mass of Two (2) Carbon Atoms
Spring Constant	1	arbitrary
Lattice Constant	3×10^{-10} m	arbitrary
No. of Wavevector Grid Points	50	-

Table 7: System Parameters for Fig. (10)

4.2 Discussion

Two dispersion values can be found for every wavevector value k in the diatomic linear chain since every unit cell in the diatomic linear chain has two degrees of freedom.

5 Problem 5: Drude Model

5.1 Results

5.1.1 Current Density vs. Electric Field Strength

The relationship between absolute current density and electric field strength is shown in Fig. (11). This relationship is linear.

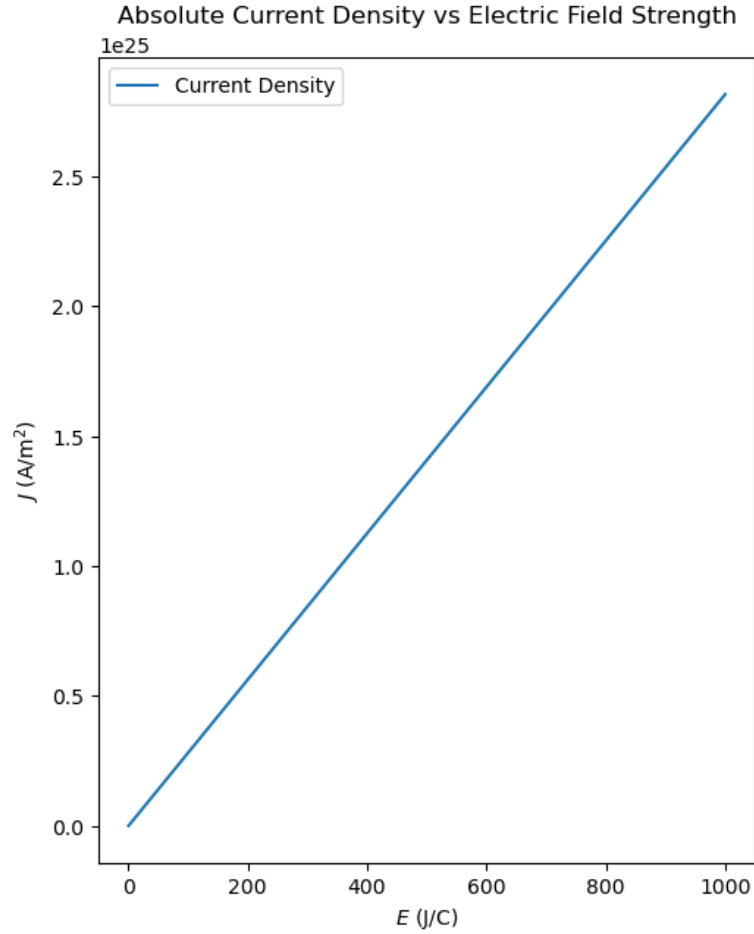


Figure 11: Absolute Current Density vs Electric Field Strength

System Parameter	Value	Notes
Carrier Density	1×10^{30} kg	arbitrary
Carrier Charge	1.602×10^{-19} kg	Charge of One (1) Electron
Effective Mass	9.109×10^{-31} kg	Mass of One (1) Electron
Relaxation Time	1 s	arbitrary

Table 8: System Parameters for Fig. (11)

5.1.2 Resistivity vs. Relaxation Time

The relationship between resistivity and relaxation time is shown in Fig. (12). They are inversely proportional to each other.

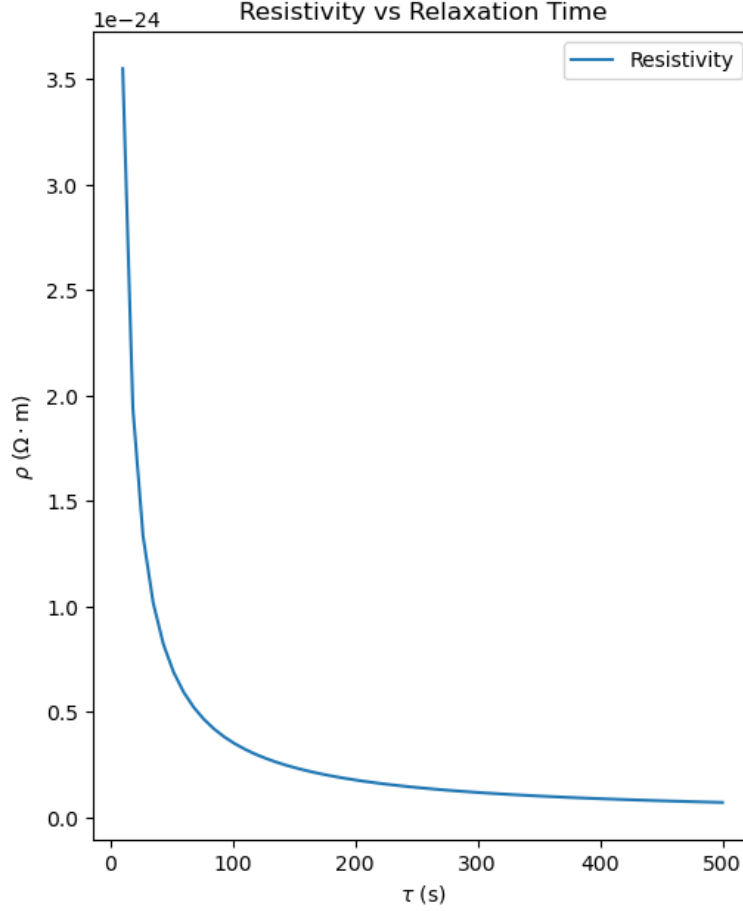


Figure 12: Resistivity vs Relaxation Time

System Parameter	Value	Notes
Carrier Density	1×10^{30} kg	arbitrary
Carrier Charge	1.602×10^{-19} kg	Charge of One (1) Electron
Effective Mass	9.109×10^{-31} kg	Mass of One (1) Electron

Table 9: System Parameters for Fig. (12)

5.2 Discussion

A drift velocity $\mathbf{v}_d$ for charge carriers in the Drude model of a metal can be defined since the charge carriers are prohibited from accelerating indefinitely by collisions with the metal host atoms.

6 Problem 6: Sommerfeld Model

6.1 Results

6.1.1 Numerical and Analytical Heat Capacity

The numerical and analytical Sommerfeld heat capacities are shown in Fig. (13). Across the temperature range displayed, there is obvious disagreement between the numerical and analytical heat capacities.

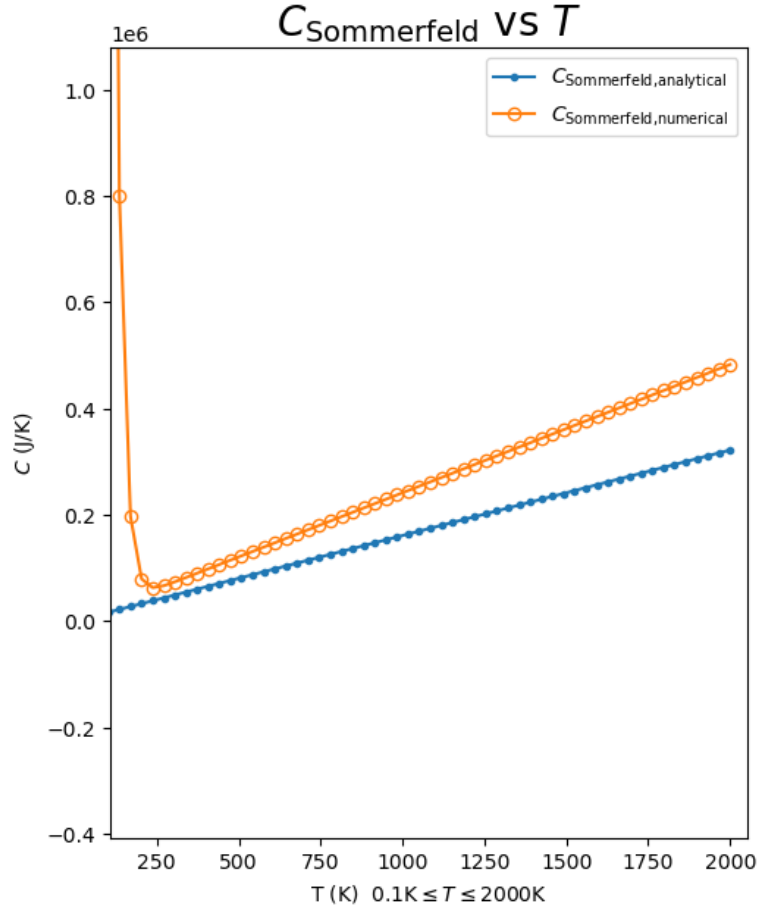


Figure 13: Numerical and Analytical Sommerfeld Heat Capacity vs Temperature

System Parameter	Value	Notes
Lower Integral Limit	1 J	-
Upper Integral Limit	1×10^{-17} J	arbitrary
No. of Energy Grid Points	2000	arbitrary
Finite Difference Magnitude (ΔT)	0.001 K	arbitrary

Table 10: System Parameters for Fig. (13)

6.2 Discussion

The Sommerfeld model gives a linear relationship between heat capacity and temperature that is valid at low temperatures. The Sommerfeld model shows that this linear relationship may be linked to free electrons obeying Fermi-Dirac statistics in the material.

References

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- [2] S. H. Simon, *The Oxford Solid State Basics*, 1st ed. Oxford University Press, 2013.